

Confidential

产品开发更改

Doc. Approval: RBCD/EIS

Effective date: 20250610

DC No. / 开发更改编号: PDECR_0016

Date / 日期: 6/19/2025

Step 1: Change request & Basic information

Customer project Name/ 客户项目名称:	57cc e-compressor	MCR No. / MCR号:	BM-00095101_003
Affect Product No. / 产品号:		Component No. / 零部件号:	
所有57cc e-compressor成品		F01Z7A008T-01& 见如下:	
Sample type / 样品类型:	☐ A ☐ B ☒ C ☒ FD	Initiator/发起人:	Fan Change (RBCD/EIS4)

Reason of changes / 更改理由:

1.将POE(RL68H)冰熊冷冻油加入BOM,纳入直采范围 (详见: 2025/5/6 e-Compressor SEC core meeting)

2.统一相关文件中的冷冻油的名称和相关铭牌定义形式

3.BOM 中data sheet 位置号删除, 统一装配图纸和优化BOM 形式。

4: 更新Qingling F01Z7P0014-02 的注油量

Customer request / 客户要求 ☐	Supplier request / 供应商请求 ☐	Design optimization / 设计优化 ☐
Correction / 更正 (错误) ☐	Process / 工艺 ☐	Other / 其它 ☒
		BOM优化 ☒

Step 2: Change proposal & Change discription

Before Change

1: BOM

无冷冻油物料直采要求。

After Change

更新后: Bom 中增加冷冻油物料, 及其对应的加注量。
冷冻油加注量将与BOM一致最终体现在铭牌上。

396.0	230.000ML	F01Z.7A0.08T-01	Oil; Refrigerant:EMDAPATE POE(RL68H)
300.1	1,000 PC	F01Z.7A0.04N-01	Wire; Grounding wire
300.2	4,000 PC	F01Z.7A0.05E-01	Damper
927.0			IMSS ID-TBD
940.0			Nameplate
940.1			Oil; POE (RL68H) 230ml
940.2			Refrigerant: R134a

如下内容为文档更新

2: 标准版Offer drawing 上定义冷冻油型号和加注量参见TCD
(Geely 提出特殊需求, 将冷冻油型号和加注量定义在图纸上)

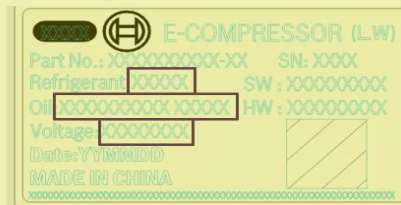
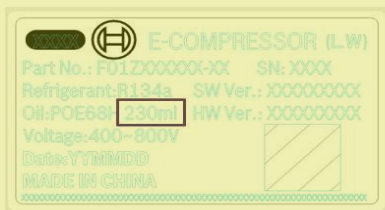
Ambient templateure and use temperature refer to TCD 压缩机环境温度和使用温度范围参见TCD
Type of refrigerant refer to TCD 制冷剂类型参见TCD
Type of refrigerant oil and filling quantity refer to TCD 冷冻油型号和加注量参见TCD

Geely offer drawing 统一成POE(RL68H)

Geely

Type of refrigerant oil and filling quantity:POE68H 230ml
冷冻油型号和加注量: POE68H 230ml

3:铭牌定义如下:

铭牌及其对应的offer drawing 统一成POE(RL68H)
仅示意 (所有涉及料号均需更新)

- 由于不同客户之间注油量不同, 具体项目参考Bom: 940.1

940.1

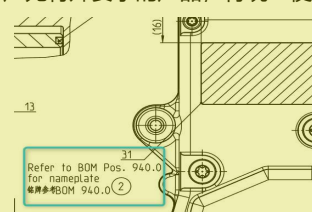
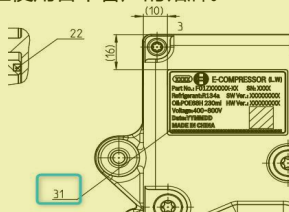
Refrigerant Oil: 230ml

- 为避免铭牌变更带来的客户图纸不断变化。冷冻油/冷媒/电压均参考BOM
- 客户图纸签署时, 提供单独的铭牌示意给客户签字

940.1
940.2
940.3Oil: POE (RL68H) 230ml
Refrigerant: R134a
Voltage: 400V~800V

4: 装配图上使用各个客户的铭牌。

客户无特殊要求的产品, 将统一使用一致的装配图



变更理由: 为降低维护effort, 除客户特殊需求之外, 不同客户项目之间可使用统一的装配图。

Initiator / 发起人:	Date / 日期:
Signature / 签名:	Date / 日期:
Check / 校对工程师	Date / 日期:
Signature / 签名:	Date / 日期:
EPQ	Date / 日期:
Signature / 签名:	Date / 日期:

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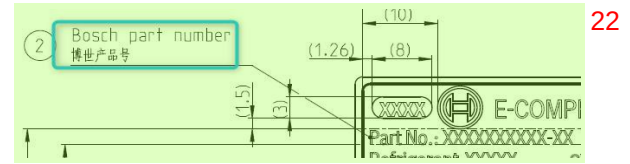
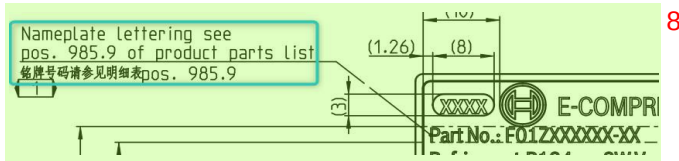
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Before Change 7

After Change 21



5: F01Z7S0003-01 功能检测规范 9

3.1.3 Test procedure/测试步骤 10

1. E-compressor code scanning/压缩机扫码 11
2. Weigh E-compressor/称重: W₀. Fill E-compressor with Refrigerant Oil Emkarate RL68H 230mL. Weigh E-compressor: W₁. 注入 230mL RL68H 冷冻油, 称重 W₁. 12
3. Assembly E-compressor to function test bench. 安装压缩机至功能台架. 13
4. Connect E-compressor inlet with air supply machine. 连接压缩机进气口与供气设备. 14
5. Connect E-compressor electrical connector. 连接电插头. 15
6. Power on and running E-compressor. 上电启动压缩机. 16
7. Monitor and record related parameters. 监控和记录相关参数. 17
8. Stop test. Weigh E-compressor after test W₂. 停机, 试验后称重 W₂. 18

统一成POE(RL68H) 并添加补油描述和注油量公差 23

3.1.3 Test procedure/测试步骤 24

1. E-compressor code scanning/压缩机扫码 25
2. Weigh E-compressor/称重: W₀. Fill E-compressor with Refrigerant Oil POE(RL68H) 230mL(±5%, density 0.98g/mL). Weigh E-compressor: W₁. 注入 POE(RL68H)230mL 冷冻油(公差±5%, 密度 0.98g/mL), 称重 W₁. 26
3. Assembly E-compressor to function test bench. 安装压缩机至功能台架. 27
4. Connect E-compressor inlet with air supply machine. 连接压缩机进气口与供气设备. 28
5. Connect E-compressor electrical connector. 连接电插头. 29
6. Power on and running E-compressor. 上电启动压缩机. 30
7. Monitor and record related parameters. 监控和记录相关参数. 31
8. Stop test. Oil replenishment to delivery status as BOM940.1 requirement. 停机, 压缩机补油至发货状态注油量要求 (参照 BOM940.1, 公差±5%, 密度 0.98g/mL). 32

6: TCD中定义冷冻油类型: 33

7: 目前BOM中992.0 位置预留了datasheet 的位置号 34

Bom 中TCD 位置号9979 35

997.0	F01Z7D0012-01	OFFER DRAWING
997.9	F01Z7S0001-01	TCD

997.0 42

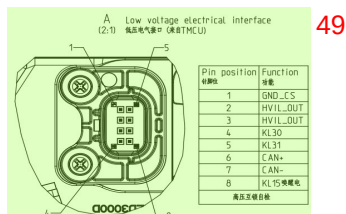
F01Z7D000H-01	OFFER DRAWING
TCD:F01Z7S0010-01	

8: offer drawing 增加追溯码备注和描述: 38

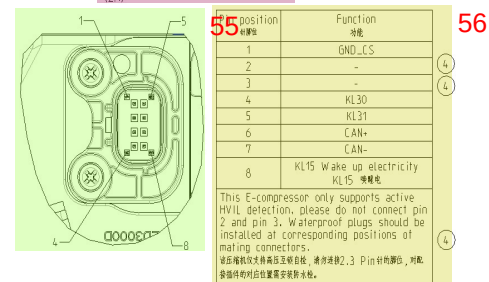
All DMC and traceability codes 所有二维码和追溯码

Except for the nameplate, all DMC and traceability codes are internally used by Bosch, the illustration is only for reference, pls refer to the actual product for accuracy and should not be subject to dimension or visual inspection criteria. 除铭牌外, 所有二维码和追溯码均为博世内部使用, 图示仅供示意, 请以产品为准, 不作为检查项目. 17

Offer drawing 电气接头信息变更 48



A Low voltage electrical interface (2:1) 低压电气接口 (乘2:1) 54



9: 更新Qingling F01Z7P0014-02 注油量 51

940.1 Refrigerant Oil: 230ml 52

940.1

Refrigerant Oil: 250mL 57

10: 高压互锁检测在ME进行,因此删除总成BOM 下的高压互锁文档 53

990.2 F01Z7S0008-01 High voltage interlock 58

PS: ME的EOL PAV文件 (ME要求的是PAV文档, 非PMB) 已经包含此项检测 59

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Step 3.1 Impact analysis / 影响分析

1. Function&Performance will be influenced? / 产品功能性能影响?		☒ yes/是	☐ no/否
If Yes, product impacton evaluation? 如果是, 产品影响评估? (如果是初次样件放行, 空白处请说明样件目的)		目前仅放行冰熊品牌POE(RL68H)冷冻油, 功能验证已完成	
Whether need sample to test? Plan? 是否需要样件进行验证? 计划?		如需放行其他品牌, 验证试验需定义。	
2.1 Interface&boundary to customer will be influenced? / 接口&边界是否影响客户?		☐ yes/是	☒ no/否
Please explain the impact and whether it has been confirmed by the customer?请说明影响和是否得到客户的确认?		对客户接口无影响, 对客户铭牌和图纸有影响, 需通知客户	
2.2 Interface&boundary to internal will be influenced? / 接口&边界是否影响内部?		☐ yes/是	☒ no/否
If Yes, internal impacton evaluation? 如果是, 内部影响是否评估? (例如: 测试接头, 噪音测试夹具等)		对内部接口无影响	
3. Mechanical strength&Durability will be influenced? / 产品可靠性、鲁棒性影响?		☒ yes/是	☐ no/否
If Yes, How to evaluate? Plan? 如果是, 请说明影响和如何评估、计划? (例如:仿真分析等)		使用冰熊品牌POE(RL68H)冷冻油的耐久试验进行中, 暂无问题反馈。	
Whether need new sample or fixture to test? Plan? 是否需要样件以及定制/改制夹具进行验证? 计划?		如需放行其他品牌, 耐久验证需定义。	
4. Others component will be influenced? / 影响其它零部件?		不影响其他零部件	
If yes, Which components will change in parallel? 如果影响, 哪些零部件也需要同时变更?			
5. Manufactory/assembly/testing will be influenced? / 加工、装配、测试是否影响?			
Feedback from manufacturing engineer? 生产制造装配是否影响, 反馈?			
对加工装配无影响			
Influence on project&product delivery time? 对项目&产品交货时间的影响?			
物料到位前生产下单临时采购			
Influence on manufacturing cost / 对生产成本的影响? (生产临时采购: 8桶/每桶5L 共计4500RMB, 纳入直采)			
☐ Increase/增加, 金额		☐ Decrease/降低, 金额	☒ No change/不变
6. Influence on supplier part? Feedback from Supplier/Purchasing?		☐ NA.(新零件放行,不涉及库存)	
Was this change confirmed with Purchasing (Supplier) for feasibility? Feedback from Supplier? 变更后的零件, 生产可行性供应商是否确认? 反馈?		PPS for purchase parts/签名 经确认, 无需签字	
Influence on purchasing part delivery time? 对采购件交货时间的影响?		(Below only for DCAC)	
涉及到变更的零件, 博世已出PR, 供应商还未来得及生产的订单, 是否已经通知供应商取消。		☐ no/否 ☒ yes/是	
7: Does the change influence on purchasing cost? 变更对采购成本的影响?		DCAC Mech. 签名	
☐ Increase/增加, 金额		☐ Decrease/降低, 金额	DCAC软/硬件 签名
☐ no change/不变		Quality: 签名	
8. Changed parts Stock& treatment / 需要变更零件的库存统计和处理? (只针对涉及到已经生产过的零件变更, 新放行零件和本条不相关)		☐ 涉及到已经生产过的零件变更	
RBCW仓库和RBCD仓库, 是否有库存,	☐ 无库存	☒ 有库存, 库存数量	零公里 (180L)
供应商处是否有库存,	☒ 无库存	☐ 有库存, 库存数量	
在途, 运输中是否有库存	☒ 无库存	☐ 有库存, 库存数量	
被影响到的实物库存处理指示: (设计工程师组织会议讨论)	☐ 可以继续使用完	☐ 改制后使用	☐ 报废
	☐ 改制费用	☐ 报废成本	
9. DFMA & Sample production OPL?		☒ no/否 ☐ yes/是	
DFMA 会议纪要, 请附件说明; 如果变更无需DFMA或前期PDECR已经做过DFMA, 请说明并列出之前的PD_ECR号 Any OPL not closed from last time sample production? 上次样件生产是否仍有OPL没有完成? 请说明或者附件说明		无DFMA	
10. PM alignment? (Influence on project Timeline? Cost?): 是否同意对项目时间和成本的影响; 需求样品订单数量		Design / 签名	
11. 如果涉及以下情况需TCR3 审核确认。 不涉及则		☐ N.A	
☐ 库存报废, 金额		☐ 采购成本增加, 金额	☐ 生产成本增加, 金额

Step 3.2 Quality Assurance Items / 检查项目

Validations 验证	Plan finish date 计划完成日期	Resp.person 负责人	Comments 备注
☐ Capability Studies_ CMK	N/A	N/A	N/A
☐ Capability Studies_ MSA	N/A	N/A	N/A
☐ MAE release	N/A	N/A	N/A
☐ Cleanness test	N/A	N/A	N/A
☐ QZ test	N/A	N/A	N/A
☐ 200h PDL	N/A	N/A	N/A
☐ Test program or fixture etc.	N/A	N/A	N/A

Step 3.3 Affected documents Check/ 影响文档检查

1 Interface FMEA-relevant / IFMEA:	☒ no/否 ☐ yes/是	Resp.person:	Due date:
2.Product FMEA-relevant / DFMEA:	☐ no/否 ☒ yes/是	Resp.person: Fan Change	Due date D-FMEA 更新已完成
3. Special Charecteristics relevant/PSC:	☒ no/否 ☐ yes/是	Resp.person:	Due date:
4. IMDS relevant:	☒ no/否 ☐ yes/是	Resp.person:	Due date:
5 Offer drawing relevant:	☐ no/否 ☒ yes/是	Resp.person: All designer	Due date 7/15/2025
6. TCD relevant:	☐ no/否 ☒ yes/是	Resp.person: QianJiong/ Sun Xudong	Due date 7/15/2025
7. Norm, WB, HF... relevant: WI	☐ no/否 ☒ yes/是	Resp.perso LI bing/Li Ke	Due date 7/15/2025

Step 4 Technical feasibility & Validation plan approval / 技术可行性&验证计划批准

Project Manger	e-Motor/DCAC Design SM	Function testing SM	Endurance testing SM	Quality SM	Maunufacture SM(Scroll)
Signature / 签名:	Signature / 签名:	Signature / 签名:	Signature / 签名	Signature / 签名:	Signature / 签名:
	N/A				N/A
COS SM	Purchasing SM	Sales SM	LOP SM	EIS Department Manger	Assembly/Testing Manger
Signature / 签名:	Signature / 签名:	Signature / 签名:	Signature / 签名:	Signature / 签名:	Signature / 签名:
					TCR3
					Signature / 签名:

Development Changes in PD-Phase

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Step 5 Documents release / 文档发布

1. Documents release (drawing, offer drawing, Bom, Spec.,) / 文档发布 (图纸, 客户图纸, ...)

Development confirmation/ 研发确认:

Date / 日期:

Step 6.1 Implementation check list / 导入清单

Departments	Y/N	Description	Responsible	Due date
Development	Y	Change BOMs & Drawings & Documents in POE system	Li Pingzheng	7/15/2025
	Y	Inform documents update (check work-on can met requirements)	Fan Change	7/15/2025
	Y	Update Offer drawing, TCD, D-FMEA	Wang Xiaolong	7/15/2025
	Y	Norm, WB, HF... PMB	Qian Jiong/Sun xudong	7/15/2025
	N	MoC, IMDS.	Li bing/ Li ke	7/15/2025
Manufacturing	N	Related (Production/Testing) equipement be ready on site	N/A	N/A
	N	Related (Production/Testing) program be ready	N/A	N/A
	N	Related (Production/Testing) tooling / cutting / fixture etc. be ready	N/A	N/A
	N	Old tooling / cutting / fixture disposal	N/A	N/A
	N	Old materials disposal	N/A	N/A
	N	Planner update the planning sheet	N/A	N/A
	N	Update FMEA	N/A	N/A
	N	Update CP/FC (Control Plan/Flow Chart)	N/A	N/A
	Y	Update WI/PDS (Include attachments.)	Li Ke	6/30/2025
	N	First batch Mark, Special Mark (Inside Package)	N/A	N/A
	N	First batch Mark, Special Mark (Outside Package)	N/A	N/A
	Y	Training	Li Ke	6/30/2025
COS	Y	Confirm the storage of old parts and coordinate the introduction date for new part (RM)	Meng Fanlong	
	Y	Confirm the delivery date of old parts and first delivery of new parts (FG)		
	N	Check sample orders which affected:material order of CKD	N/A	N/A
	N	Confirm production scheduling according to the alignment, any changes share the information to MOEx,MFE	N/A	N/A
	N	Confirm the old stock/ do prioritize delivery and inventory handling	N/A	N/A
	N	Inform the first delivery to PMO	N/A	N/A
PPS (PPS->PMQ/PQA)	Y	Check sample orders which affected: material order of purchasing	Zheng Xuan	
	Y	Inform internal related departments(COS,MFE,MOEx)with following	Zheng Xuan	
	Y	Update incoming inspection plan	Zheng Xuan	
QMM1	N	Update testing programe on testing equipment	N/A	N/A
	N	Update inspection plan for CKD parts	N/A	N/A
EIS4	Y	Distribute the Offer drawing, TCD to customer	All designer	N/A
LOP	Y	Check 10 digit material order	Ding Yating	
PMO	N	Check sample orders which affected: Customer order	N/A	N/A
	N	Inform Customer the first delivery information	N/A	N/A

Step 6.2 Implementation date / 执行日期

Planned implementation date:

变更计划执行日期

Step 6.3 Implementation approval / 导入清单

Development// 研发	Production/生产	COS/样品	Purchasing/采购	Quality/质量	LOP/ 物流
Signature / 签名:	Signature / 签名:	Signature / 签名:	Signature / 签名:	Signature / 签名:	Signature / 签名:

Usage list			
		Part No.	Status
1:	冷冻油加注	F01Z7A008T-01	Ongoing
2:	Geely offer drawing 更新	F01Z7D0012-01	7/15/2025
		F01Z7P0026-01	7/15/2025
3:	铭牌更新	F01Z7P008T-01	7/15/2025
		F01Z7P0065-01	7/15/2025
	Bom 更新	All e-compressor	7/15/2025
4	装配图纸更新	F01Z7P0012-01	7/15/2025
		F01Z7P0014-02	7/15/2025
5	功能规范更新	F01Z7S0003-01	7/15/2025
6	TCD 更新	All TCD	7/15/2025
7	Bom 更新	All e-compressor	7/15/2025
9	Offer drawing 更新	F01Z7D0012-01	7/15/2025
		F01Z7D0014-02	7/15/2025
9	注油量更新 BOM	F01Z7P0014-02	Done